

DroidLink Periscope Logic Lights

ESP32-C3 Super Mini controller for the Printed-Droid Periscope Logic Lights.

This firmware keeps the original Periscope lighting effects and adds DroidLink control over ESP-NOW.

The Periscope can still be controlled directly from USB Serial, but normal operation is through the DroidLink Master.

How DroidLink Control Works

The communication path is:

```
Display / RC / Master Sequence
    ↓
DroidLink Master
    ↓
ESP-NOW
    ↓
DroidLink Periscope
    ↓
Periscope LED Effects
```

The DroidLink command prefix for the Periscope is:

```
:PS
```

Examples:

```
:PSOFF
```

Turns the Periscope lights off.

```
:PSQ1
```

Runs Periscope Sequence 1.

```
:PSM185
```

Runs Main LED Effect 1, White, at Speed 5.

The Periscope firmware removes the `:PS` prefix before sending the remaining command to the original Periscope lighting engine.

Example:

```
DroidLink command: :PSQ4
Periscope command: Q4
Result: Police Lights sequence
```

First-Time DroidLink Setup

The Periscope and Master must know each other's MAC addresses.

Step 1 — Install the Periscope Firmware

Install the DroidLink Periscope firmware using the DroidLink Installer.

After installation, open the installer console.

The Periscope will display its own ESP-NOW MAC address:

```
DroidLink Periscope – Boot

Device MAC: E4:B3:23:C6:95:00
```

Your actual MAC address will be different.

Copy this address.

Step 2 — Add the Periscope to the Master

Open the DroidLink Master configuration.

Add the Periscope MAC address to an available slave slot.

The current Periscope firmware identifies itself as:

```
Node ID: 6
Role: Periscope
Device: Periscope
```

Save the Master configuration and power or reboot the Master.

Step 3 — Enter the Master MAC

On first boot, the Periscope will stop and display:

```
=====
FIRST TIME SETUP
=====

Paste Master MAC like:
9C:13:9E:A8:6F:A8
```

Paste the DroidLink Master MAC address into the console.

The Periscope will confirm the value:

```
Received MAC: 9C:13:9E:A8:6F:A8
```

It will then save the MAC and reboot:

```
Master MAC saved!
Rebooting...
```

The Master MAC is stored in ESP32 Preferences and remains saved after power loss.

Successful Connection

When configuration is correct, the Periscope boot output will include:

```
Master peer added
ESP-NOW ready
```

When the Master sends a valid packet, the Periscope reports:

```
Master Seen: 1
ESP-NOW RX: :PSOFF
LIGHT CMD: OFF
```

The Master will also discover the Periscope after broadcasting:

```
:CAP?
```

The Master's console should show:

```
-----  
Capability received  
-----
```

```
Node: Slave (ID 6)
```

```
Roles: Periscope
```

```
Adapters: Periscope
```

The `:CAP?` command is a DroidLink system command. It is handled internally and is not a Periscope lighting effect.

Changing an Incorrect Master MAC

If the wrong Master MAC was entered, connect the Periscope to USB and open a serial terminal.

Type:

```
NEWMAC
```

The Periscope will erase the saved Master MAC:

```
Clearing stored Master MAC...  
Rebooting...
```

After reboot, the first-time setup prompt will return.

```
Paste Master MAC like:  
9C:13:9E:A8:6F:A8
```

Enter the correct Master MAC.

Master Communication Check

After boot, the Periscope waits for a valid packet from the configured Master.

If no valid Master packet is received within approximately 15 seconds, the console displays:

```
No Master communication detected.  
Type NEWMAC to reconfigure.
```

This can mean:

- The Master is not powered on
- The wrong Master MAC was entered
- The Periscope MAC was not added to the Master
- The Master and Periscope are not using the same ESP-NOW channel
- The Master has not sent a command yet

Receiving any valid packet from the configured Master changes:

```
Master Seen: 0
```

to:

```
Master Seen: 1
```

DroidLink Command Format

All DroidLink Periscope commands begin with:

```
:PS
```

The complete format is:

```
:PS[PERISCOPE COMMAND]
```

Examples:

DroidLink Command	Periscope Receives	Description
:PSON	ON	Turn the lights on
:PSOFF	OFF	Turn the lights off
:PSX	X	Emergency off
:PSQ1	Q1	Party Mode
:PSQ4	Q4	Police Lights
:PSQ6	Q6	Knight Rider
:PSM185	M185	Main LEDs, white pulse
:PST647	T647	Top LEDs, blue chase
:PSA199	A199	All LEDs, pink fast

Basic Commands

These are the native Periscope commands.

When using DroidLink, add `:PS` before the command.

Native Command	DroidLink Command	Description
ON	:PSON	Turn LEDs on
OFF	:PSOFF	Turn LEDs off
X	:PSX	Emergency off
?	:PS?	Show status/help when sent through Serial

Sequence Commands

A sequence command selects and configures the requested lighting sequence. If the Periscope LEDs are currently off, the sequence is prepared but will not be displayed until the lights are enabled with:

```
:PSON
```

For example, to select and display the Classic R2 Startup sequence:

:PSQ0
:PS0N

If the LEDs are already on, sending another sequence command changes to that sequence immediately. Use :PSOFF or :PSX to turn the LEDs off.

Native Command	DroidLink Command	Description
Q0	:PSQ0	Classic R2 Startup
Q1	:PSQ1	Party Mode
Q2	:PSQ2	Bright Pulse
Q3	:PSQ3	Communication Mode
Q4	:PSQ4	Police Lights
Q5	:PSQ5	Red Alert
Q6	:PSQ6	Knight Rider
Q7	:PSQ7	Searchlight
Q8	:PSQ8	Stealth Mode
Q9	:PSQ9	Diving
Q10	:PSQ10	Surfacing
Q11	:PSQ11	Calm Blue
Q12	:PSQ12	System Boot
Q13	:PSQ13	Fire Mode
Q14	:PSQ14	Celebration
Q15	:PSQ15	Charging
Q16	:PSQ16	Hyperdrive
Q17	:PSQ17	Malfunction
Q18	:PSQ18	Scan Complete
Q19	:PSQ19	Sonar Ping
Q20	:PSQ20	Auto Demo Mode

LED Groups

Code	Group
M	Main LEDs
T	Top LEDs
B	Bottom LEDs
S	Both sides
L	Left side
R	Right side
K	Back LEDs
A	All LEDs

Custom Command Format

The native Periscope format is:

```
[GROUP][EFFECT][COLOR][SPEED]
```

The DroidLink format is:

```
:PS[GROUP][EFFECT][COLOR][SPEED]
```

Example:

```
:PSM185
```

Meaning:

- :PS = Route to the Periscope Slave
- M = Main LEDs
- 1 = Effect 1
- 8 = White
- 5 = Medium speed

The Periscope receives:

```
M185
```


Colors

Number	Color
0	Red
1	Yellow
2	Green
3	Cyan
4	Blue
5	Magenta
6	Orange
7	Purple
8	White
9	Pink

Speed

Number	Speed
0	Slowest
5	Medium
9	Fastest

Values from 0 through 9 may be used.

Useful DroidLink Examples

Turn Everything Off

```
:PSOFF
```

Party Mode

:PSQ1

Police Lights

:PSQ4

Knight Rider

:PSQ6

Main LEDs — White Pulse

:PSM185

Top LEDs — Blue Chase

:PST647

All LEDs — Pink Fast

:PSA199

Auto Demo

:PSQ20

Direct USB Serial Control

The Periscope can still be controlled directly through USB Serial.

When using Serial directly, do not add the DroidLink `:PS` prefix.

Use:

```
Q1
```

instead of:

```
:PSQ1
```

Use:

```
M185
```

instead of:

```
:PSM185
```

The main firmware serial handler also accepts:

```
NEWMAC
```

to erase the saved Master MAC.

Serial Settings

- Baud Rate: `9600`
- Line Ending: `Newline` or `Both NL & CR`

Commands are converted to uppercase by the firmware, so lowercase input is also accepted.

Hardware

Board

- ESP32-C3 Super Mini

LED Pins

Function	GPIO
Main	GPIO5
Right	GPIO7
Left	GPIO4
Bottom	GPIO3
Top	GPIO6
Back	GPIO10

ESP-NOW Settings

The Periscope currently uses:

```
Wi-Fi Station Mode
ESP-NOW Channel 1
Encryption Disabled
```

The Master must also use ESP-NOW Channel 1.

Only packets from the saved Master MAC are accepted.

Packets from other MAC addresses are ignored.

FastLED Requirement

Use:

```
FastLED 3.9.0
```

PlatformIO configuration:

```
lib_deps =
    fastled/FastLED@3.9.0
```

Do not use the caret version:

```
fastled/FastLED@^3.9.0
```

Pinning the exact version prevents PlatformIO from automatically installing a newer version.

Newer FastLED versions may cause:

- Flickering
- Incorrect colors
- Timing problems
- ESP32 RMT conflicts
- Unstable multi-strip behavior

Current DroidLink Identity

The current firmware reports:

```
Node ID: 6
Role: ROLE_PERISCOPE
Device: DEV_PERISCOPE
```

The Master routes commands beginning with:

```
:PS
```

to the node advertising:

```
ROLE_PERISCOPE
```

The Periscope therefore does not need its MAC hardcoded into command-routing logic after capability discovery. The Master routes using the advertised Periscope role.

Quick Setup Summary

1. Install Periscope firmware
2. Copy the Periscope MAC from the installer console
3. Add the Periscope MAC to the DroidLink Master
4. Power or reboot the Master
5. Paste the Master MAC into the Periscope console
6. Periscope saves the MAC and reboots
7. Master sends :CAP?
8. Periscope reports Role: Periscope
9. Send commands using :PS

Example:

```
:PSQ4
```

Runs Police Lights.

```
:PSOFF
```

Turns the Periscope lights off.

Viewing This README in VS Code

Open `README.md` and press:

```
CTRL + SHIFT + V
```

Or right-click inside the file and select:

```
Open Preview
```